3.1 Introduction to Determinants

Notation: A_{ij} the matrix obtained from matrix A by deleting the ith row and the jth column of A.

Example: $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$ $A_{23} =$

Recall that $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$ and we let $\det[a] = a$.

For $n \geq 2$, the determinant of an n x n matrix $A = [a_{ij}]$ is given by

$$\det A = a_{11}\det A_{11} - a_{12}\det A_{2} + \cdots + (-1)^{1+n} a_{1n}\det A_{1n} = \sum_{j=1}^{n} (-1)^{1+j} a_{1j}\det A_{j}$$

Ex 1: Compute the determinant of $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$

Common notation: $\det\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} = \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix}$

So $\begin{vmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 1 \end{vmatrix} = 1\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} + 0\begin{vmatrix} 3 & -1 \\ 2 & 0 \end{vmatrix}$

The (i, j) – cofactor of A is the number C_{ij} where $C_{ij} = (-1)^{i+j}\det A_{ij}$

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 1 \end{vmatrix} = 1C_{11} + 2C_{12} + 0C_{13}$$ (cofactor expansion across row 1)

Theorem 1: The determinant of an n x n matrix A can be computed by a cofactor expansion across any row or down any column:

$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$ (expansion down column j)

$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$ (expansion across row i)

Use a matrix of signs to determine $(-1)^{i+j}$

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Ex 2: Compute the determinant of $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$ using cofactor expansion down column 3.

Ex 3: Compute the determinant of $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 5 \end{bmatrix}$

Triangular matrices**:**

$\begin{bmatrix} * & * & \cdots & * & * \\ 0 & * & \cdots & * & * \\ 0 & 0 & \ddots & * & * \\ 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * \end{bmatrix}$ Upper triangular

$\begin{bmatrix} * & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 \\ * & * & \ddots & 0 & 0 \\ * & * & \cdots & * & 0 \\ * & * & \cdots & * & * \end{bmatrix}$ lower triangular

Theorem 2: If A is a triangular matrix, then det A is the product of the main diagonal entries of A.

Ex 4: Evaluate $\begin{vmatrix} 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -3 & 5 \\ 0 & 0 & 0 & 4 \end{vmatrix}$

3.2 Properties of Determinants

Theorem 3: Let A be a square matrix.

a. If a multiple of one row of A is added to another row of A to produce a matrix B, then det A = det B.
b. If two rows of A are interchanged to produce B, then det B = - det A.
c. If one row of A is multiplied by k to produce B, then det $B = k$ (det A).

Ex 1: Compute by row reduction techniques to make it into a triangular matrix. $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 0 & 0 \\ 2 & 7 & 6 & 10 \\ 2 & 9 & 7 & 11 \end{vmatrix}$

Theorem 3(c) indicates that $\begin{vmatrix} * & * & * \\ -2k & 5k & 4k \\ * & * & * \end{vmatrix} = k \begin{vmatrix} * & * & * \\ -2 & 5 & 4 \\ * & * & * \end{vmatrix}$

Ex 2: Compute by row reduction techniques to make it into a triangular matrix $\begin{vmatrix} 2 & 4 & 6 \\ 5 & 6 & 7 \\ 7 & 6 & 10 \end{vmatrix}$

Suppose A has been reduced to an echelon form U by row replacements and row interchanges, then

$$\det A = \begin{cases} (-1)^r \,(\text{product of pivots in U}) & \textit{when } A \textit{ is invertible} \\ 0 & \textit{when } A \textit{ is not invertible} \end{cases}$$

Theorem 4: A square matrix is invertible if and only if $\det A \neq 0$.

Theorem 5: If A is an n x n matrix, then $\det A^T = \det A$

Implications of Theorem 5?

Theorem 3 still holds if the word row is replaced with_____.

Theorem 6: (Multiplicative Property) For n x n matrices A and B, det (AB) = (det$A$)(det$B$)

Ex: Compute $\det A^3$ if det A = 5.

Ex: For n x n matrices A and B, show that A is singular if $\det B \neq 0$ and det AB = 0.

Ex: Find the determinants of the following elementary matrices by inspection:

a. $\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$ b. $\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$ c. $\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -9 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$

Ex.: Given that $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$, find

a. $\begin{vmatrix} d & e & f \\ g & h & i \\ a & b & c \end{vmatrix}$

b. $\begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix}$

c. $\begin{vmatrix} a+g & b+h & c+i \\ d & e & f \\ g & h & i \end{vmatrix}$

d. $\begin{vmatrix} -3a & -3b & -3c \\ d & e & f \\ g-4d & h-4e & i-4f \end{vmatrix}$

3.3 Cramer's Rule, Volume, And Linear Transformations

Cramer's Rule

- Can be used to study how the solution of A**x** = **b** is affected by changes in the entries of **b**.

Theorem 7 – Cramer's Rule.

Let A be an invertible n x n matrix. For any b in R^n, the unique solution x of Ax = b has entries given by

$$x_i = \frac{\det A_i(b)}{\det A}, \; i = 1, 2, \cdots, n.$$

Ex 1: Use Cramer's Rule to solve $\begin{aligned} 4x_1 + x_2 &= 6 \\ 5x_1 + 2x_2 &= 7 \end{aligned}$

Ex 2: Use Cramer's Rule to solve

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 4 \\ -x_1 \quad\quad + 2x_3 &= 2 \\ 3x_1 + x_2 + 3x_3 &= -2 \end{aligned}$$

Application to Engineering

- A number of important engineering problems, particularly in electrical engineering and control theory, can be analyzed by Laplace transforms.

- This approach converts an appropriate system of linear differential equations into a system of linear algebraic equations whose coefficients involve a parameter.

In Ex 3 and Ex 4, determine the values of the parameter s for which the system has a unique solution, and describe the solution.

Ex 3:

$$\begin{aligned} 3sx_1 - 5x_2 &= 3 \\ 9x_1 + 5sx_2 &= 2 \end{aligned}$$

Ex 4:

$$\begin{aligned} 2sx_1 + x_2 &= 1 \\ 3sx_1 + 6sx_2 &= 2 \end{aligned}$$

A Formula for A^{-1} (Read Page 203)

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

Definition: The matrix of cofactors is called the adjugate (or **classical adjoint**) of A, denoted by **adj A**.

Theorem 8 – An Inverse Formula

Let A be an invertible n x n matrix. Then $A^{-1} = \frac{1}{\det A}\ adjA$.

In Ex 5 and 6, compute the adjugate of the given matrix, and then use Theorem 8 to give the inverse of the matrix.

Ex 5: $\begin{bmatrix} 1 & 1 & 3 \\ 2 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

Ex 6: $\begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$

Determinants as Area or Volume

Theorem 9

If A is a 2 x 2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$.

If A is a 3 x 3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.

(Read pgs 205 – 206 for the proof)

Ex 7: Find the area of the parallelogram whose vertices are (0, 0), (-1, 3), (4, -5), (3, -2)

Ex 8: Find the area of the parallelogram whose vertices are (0, -2), (6, -1), (-3, 1), (3, 2).

Ex 9: Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at (1, 4, 0), (-2, -5, 2), and (-1, 2, -1).

Linear Transformations

- Determinants can be used to describe an important geometric property of linear transformations in the plane and in R^3.

- If T is a linear transformation and S is a set in the domain of T, let T(S) denote the set of images of points in S.

- We are interested in how the area (or volume) of T(S) compares with the area (or volume) of the original set S. For convenience, when S is a region bounded by a parallelogram, we refer to S as a parallelogram.

Theorem 10:

Let $T : R^2 \to R^2$ be the linear transformation determined by a 2 x 2 matrix A. If S is a parallelogram in R^2, then $\{\text{area of T(s)}\} = |\det A| \bullet \{\text{area of S}\}$.

If T is determined by a 3 x 3 matrix A, and if S is a parallelepiped in R^3, then
$\{\text{volume of T(s)}\} = |\det A| \bullet \{\text{volume of S}\}$.

- The conclusions of Theorem 10 hold whenever S is a region in R^2 with finite area or a region in R^3 with finite volume.

Ex 10: Let S be the parallelogram determined by the vectors $b_1 = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and let $A = \begin{bmatrix} 7 & 2 \\ 1 & 1 \end{bmatrix}$.
Compute the area of the image of S under the mapping $x \mapsto Ax$.